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A IMPLEMENTATION DETAILS

A.1 Implementation details: LLM-based WOK recommender

We used OpenAI's `gpt-3.5-turbo-0613` ("gpt-3.5" in the main text) and `gpt-4-1106-preview` ("gpt-4" in the main text) APIs, as well as Anthropic's `claude-instant-1.2` ("claude-v1" in the main text) and `claude-2.1` ("claude-v2.1" in the main text) APIs. All LLMs used a temperature parameter of 0.0, a *top-p* parameter of 1 (default for both OpenAI and Anthropic APIs), and a *top-k* parameter of 250 (default). The base prompt template used for all experiments is given in Code Listing 1.

```
"""
You are a helpful movie-expert AI tasked with recommending a collection of movies based on a
    user's watch history. The user has watched the following movies in the past:
{watch_history}

# Output instructions
- Immediately start with the movies. Do not provide an introduction.
- Provide a list of {nr_items} movies.
- For each movie, start a new line, indicate the position in the movie list (that is, 1., 2.,
     ...).
- Name the title of the movie (without quotation marks!) and then in parentheses the release
    year.
- Do not recommend movies that the user has already watched. Those are the ones listed above.
- Do not recommend movies that are newer than 2008.

Now create the movie list!"""
```

Code Listing 1: Prompt template used for the LLM movie recommender. The placeholder `watch_history` is replaced by a list of movies watched by the user at runtime.

B ON THE DIFFERENCE OF AVERAGE POPULARITY LIFT AND LOG POPULARITY DIFFERENCE AND THE IMPORTANCE OF STATISTICAL WELL-BEHAVEDNESS (DESIDERATUM 1).

We have seen that average popularity lift and log popularity difference are strongly correlated (Section 4.6), but differ in that average popularity lift does not satisfy the Desiderata (1), (3) and (4), see Table 1. We have motivated the importance of these desiderata theoretically in Section 3.1, and investigate their practical implications here, specifically for well-behavedness (1). Figure 5 illustrates the cumulative averages of popularities $\sum_{i+1}^{N} \phi(a_i)/N$ for N a varying number of observations (x-axis) under average popularity lift (left) and log popularity difference (right). For the average popularity lift metric, there is a sudden jump when the largest popularity item is added (around $i = 7500$). This results from the Power-law distribution of unnormalized scores (g =identity, Table 1) and the resulting ill-behavedness. As a consequence, average popularity lift measurements will be highly sensitive to whether or not individual, high popularity items are included in recommendation or user sets p_u or p_r, respectively, and will measure any differences in the low-mid popularity regime less reliably.

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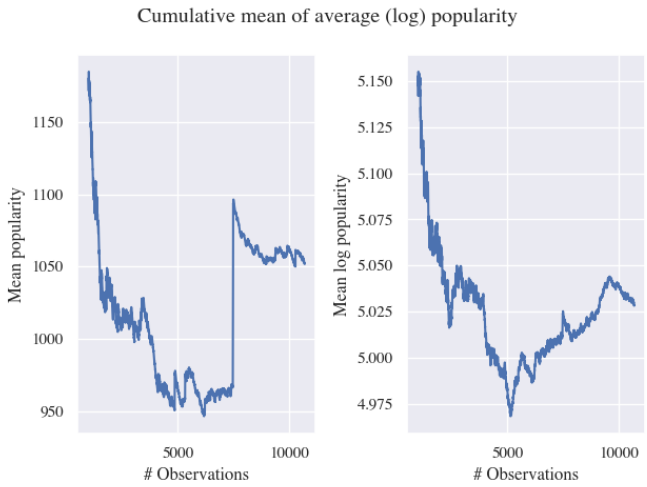

Figure 5: Behavior of the average popularity bias as a function of the data points.